

The Astrophysical S-factor

Coulomb Barrier Penetration, the Sommerfeld Parameter, the Gamow Peak, and Applications to Stellar Burning

Dilruwan DD

Based on A.C. Phillips, *The Physics of Stars*, Ch. 4

June 26, 2026

Outline

- 1 Motivation
- 2 Barrier Penetration: WKB Derivation
- 3 The Astrophysical S-factor
- 4 Thermonuclear Reaction Rates and the Gamow Peak
- 5 Applications
- 6 Summary

Why fusion needs quantum mechanics

- Two nuclei of charge Z_A, Z_B feel a repulsive Coulomb potential

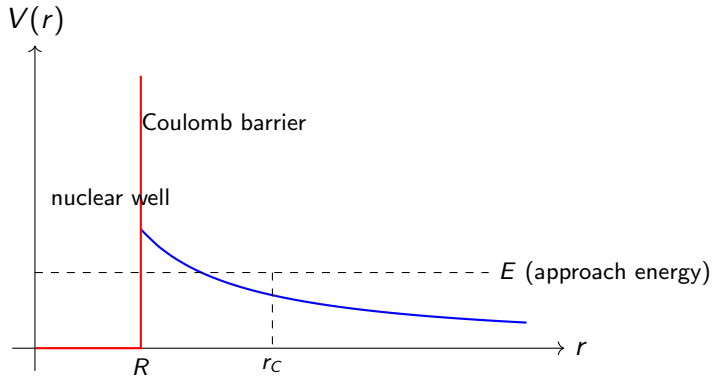
$$V(r) = \frac{Z_A Z_B e^2}{4\pi\epsilon_0 r}, \quad r > R \text{ (nuclear radius).}$$

- Classical condition for fusion: kinetic energy E must exceed the barrier height

$$E_C = \frac{Z_A Z_B e^2}{4\pi\epsilon_0 R} \approx 1.4 \frac{Z_A Z_B}{R(\text{fm})} \text{ MeV.}$$

- At $T \sim 10^7$ K, $kT \sim 1$ keV $\ll E_C \sim$ MeV.
- Classically, the Boltzmann suppression $\exp(-E_C/kT) \sim e^{-1000}$ — fusion should never happen.
- **Resolution:** quantum tunnelling lets nuclei react at energies *far below* E_C .

The potential landscape



- Classically forbidden region: $R < r < r_C$, where $V(r) > E$.
- Quantum mechanically, the wavefunction does not vanish there — it decays exponentially and *leaks through*.

Setting up the problem

Relative motion of the two nuclei (reduced mass $m_r = m_A m_B / (m_A + m_B)$) obeys

$$\left[-\frac{\hbar^2}{2m_r} \nabla^2 + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}).$$

For s -wave (zero angular momentum) penetration of a barrier, the WKB transmission factor is

$$P(E) = \exp[-G(E)], \quad G(E) = \frac{2}{\hbar} \int_R^{r_C} \sqrt{2m_r (V(r) - E)} \, dr,$$

where the classical turning point r_C satisfies $V(r_C) = E$, i.e.

$$r_C = \frac{Z_A Z_B e^2}{4\pi\epsilon_0 E}.$$

Evaluating the Gamow integral (1)

Substitute $r = r_C \cos^2 \theta$. Then

$$V(r) - E = E \left(\frac{r_C}{r} - 1 \right) = E \tan^2 \theta, \quad dr = -2r_C \cos \theta \sin \theta d\theta.$$

The turning points $r = R \rightarrow \theta = \theta_0 = \arccos \sqrt{R/r_C}$ and $r = r_C \rightarrow \theta = 0$.

The integrand becomes

$$\sqrt{2m_r(V - E)} |dr| = 2r_C \sqrt{2m_r E} \sin^2 \theta d\theta.$$

$$\Rightarrow G(E) = \frac{4r_C \sqrt{2m_r E}}{\hbar} \int_0^{\theta_0} \sin^2 \theta d\theta.$$

Evaluating the Gamow integral (2): the Sommerfeld parameter

Define the **Sommerfeld parameter**

$$\boxed{\eta \equiv \frac{Z_A Z_B e^2}{4\pi\epsilon_0 \hbar v}}, \quad v = \sqrt{\frac{2E}{m_r}} \text{ (relative speed).}$$

One can show directly that

$$\frac{r_C \sqrt{2m_r E}}{\hbar} = 2\eta.$$

Carrying out $\int_0^{\theta_0} \sin^2 \theta \, d\theta = \frac{1}{2}\theta_0 - \frac{1}{4}\sin 2\theta_0$ gives the *exact* WKB result

$$\boxed{G(E) = 4\eta \left[\arccos \sqrt{x} - \sqrt{x(1-x)} \right]}, \quad x \equiv \frac{R}{r_C} \ll 1.$$

The thick-barrier (Gamow) limit

For $R \ll r_C$ (i.e. $x \rightarrow 0$), use $\arccos \sqrt{x} \approx \frac{\pi}{2} - \sqrt{x}$:

$$G(E) \approx 2\pi\eta - 8\eta\sqrt{x} = 2\pi\eta - 8\sqrt{\eta^2 R/r_C}.$$

The first term dominates ($R/r_C \sim 10^{-4}$ or smaller in stars), so

$$P(E) = \exp[-G(E)] \approx \exp(-2\pi\eta).$$

Physical content: $\eta \propto Z_A Z_B / v$, so the tunnelling suppression grows with charge and falls with relative speed (energy) — exactly the competition that produces a finite, calculable fusion rate.

The Gamow energy

It is conventional to write the exponent in terms of an energy scale, the **Gamow energy** E_G , defined through

$$2\pi\eta \equiv \sqrt{\frac{E_G}{E}} \quad \Rightarrow \quad \boxed{E_G = (\pi\alpha Z_A Z_B)^2 2m_r c^2}, \quad \alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137}.$$

(Check: $(2\pi\eta)^2 = Z_A^2 Z_B^2 e^4 m_r / (8\epsilon_0^2 \hbar^2 E) = E_G / E$ — consistent.)

So the barrier-penetration probability takes the compact form

$$\boxed{P(E) \approx \exp\left[-\sqrt{\frac{E_G}{E}}\right]}.$$

Example: $p + p$: $E_G = 493$ keV. At $kT = 1$ keV ($T \approx 10^7$ K),
 $P \sim \exp[-(493)^{1/2}] \approx \exp(-22).$

From penetration probability to cross-section

A nuclear cross-section can be written as a product of three factors:

$$\sigma(E) \propto \underbrace{\frac{1}{E}}_{\substack{\text{geometric} \\ (\propto \lambda^2 = h^2/2m_r E)}} \times \underbrace{\exp\left[-\sqrt{E_G/E}\right]}_{\text{Coulomb penetration}} \times \underbrace{|M|^2}_{\text{nuclear matrix element}}.$$

- The $1/E$ comes from the de Broglie wavelength of relative motion, $\lambda^2 = h^2/p^2 = h^2/(2m_r E)$.
- The exponential is the Gamow penetration factor just derived.
- All remaining (slowly-varying) nuclear physics is collected into one function.

Definition of the S-factor

$$\sigma(E) = \frac{S(E)}{E} \exp \left[- \left(\frac{E_G}{E} \right)^{1/2} \right]$$

- $S(E)$ is the **astrophysical S-factor**: it isolates the genuinely nuclear (strong/EM/weak) physics from the steeply varying, purely kinematic/Coulomb behaviour.
- Away from resonances, $S(E)$ is a smooth, slowly varying function of E — ideal for extrapolation.
- Near an isolated narrow resonance, $S(E)$ acquires a Breit–Wigner peak.
- $S(E)$ is small for reactions proceeding via the *weak* interaction (e.g. $p + p \rightarrow d + e^+ + \nu_e$, $S_{pp}(0) = 3.8 \times 10^{-22}$ keV b), larger for electromagnetic capture (e.g. $p + d \rightarrow {}^3\text{He} + \gamma$, $S_{pd}(0) = 2.5 \times 10^{-4}$ keV b), and largest for strong-interaction-dominated reactions.

Why the S-factor is useful

- Laboratory accelerators can usually only measure $\sigma(E)$ down to energies of order hundreds of keV – a few MeV (background/yield limitations).
- Stellar fusion occurs at the much lower *Gamow-peak* energy E_0 (often a few to tens of keV) – see next section.
- Because $\sigma(E)$ itself varies by tens of orders of magnitude over this range (dominated by the Gamow factor), direct measurement at E_0 is essentially impossible.
- $S(E)$ varies slowly, so it *can* be measured at accessible energies and extrapolated smoothly down to E_0 .
- Caveat: low-energy laboratory data must be corrected for **electron screening** (atomic/molecular electrons partially shield the Coulomb barrier, enhancing the measured cross-section relative to the bare-nucleus value used in stars).

Reaction rate per unit volume

For nuclei A, B with concentrations n_A, n_B and Maxwellian relative-speed distribution at temperature T ,

$$R_{AB} = n_A n_B \langle \sigma v_r \rangle, \quad \langle \sigma v_r \rangle = \left[\frac{8}{\pi m_r} \right]^{1/2} \left[\frac{1}{kT} \right]^{3/2} \int_0^\infty E \sigma(E) e^{-E/kT} dE.$$

Insert the S-factor parametrisation of $\sigma(E)$:

$$R_{AB} = n_A n_B \left[\frac{8}{\pi m_r} \right]^{1/2} \left[\frac{1}{kT} \right]^{3/2} \int_0^\infty S(E) \exp \left[-\frac{E}{kT} - \left(\frac{E_G}{E} \right)^{1/2} \right] dE.$$

The integrand contains two competing exponentials:

- $e^{-E/kT}$: Boltzmann tail — favours *low* E (fewer nuclei have to "borrow" much energy).
- $e^{-\sqrt{E_G/E}}$: barrier penetration — favours *high* E .

The Gamow peak: locating the maximum

Let $f(E) = -E/kT - (E_G/E)^{1/2}$. Setting $f'(E_0) = 0$:

$$f'(E) = -\frac{1}{kT} + \frac{1}{2} E_G^{1/2} E^{-3/2} = 0 \implies E_0^{3/2} = \frac{kT}{2} E_G^{1/2}.$$

$$E_0 = \left[\frac{E_G (kT)^2}{4} \right]^{1/3}$$

This is the **Gamow peak** energy: the energy at which fusion is most likely to occur in a thermal plasma — *not* the most probable thermal energy ($\sim kT$), and *not* the barrier height E_C , but a compromise dictated by the product of the two competing exponentials.

The width of the Gamow peak

Taylor-expand $f(E)$ about E_0 using

$$f''(E) = -\frac{3}{4}E_G^{1/2}E^{-5/2} \implies f(E) \approx f(E_0) - \frac{1}{2}|f''(E_0)|(E - E_0)^2.$$

Writing this as a Gaussian $\exp[-(E - E_0)^2/(\Delta/2)^2]$ defines the **width** of the fusion window:

$$\Delta = \frac{4}{3^{1/2}2^{1/3}} E_G^{1/6} (kT)^{5/6}.$$

$$\exp\left[-\frac{E}{kT} - \left(\frac{E_G}{E}\right)^{1/2}\right] \approx \exp\left[-3\left(\frac{E_G}{4kT}\right)^{1/3}\right] \exp\left[-\left(\frac{E - E_0}{\Delta/2}\right)^2\right].$$

Fusion is concentrated in the narrow window $E = E_0 \pm \Delta/2$.

Closed-form reaction rate

If $S(E) \approx S(E_0)$ across the (narrow) Gamow window, the integral collapses to $\sqrt{\pi} \Delta/2$ times the peak integrand value, giving

$$R_{AB} = 6.48 \times 10^{-24} \frac{n_A n_B}{A_r Z_A Z_B} S(E_0) \left(\frac{E_G}{4kT} \right)^{2/3} \exp \left[-3 \left(\frac{E_G}{4kT} \right)^{1/3} \right] \text{ m}^{-3} \text{ s}^{-1},$$

with A_r the reduced mass in amu and $S(E_0)$ in keV barns. This is the master formula of non-resonant thermonuclear reaction-rate theory.

Effective temperature exponent. Differentiating the dominant exponential gives

$$\frac{dR_{AB}}{dT} \approx \left[\frac{E_G}{4kT} \right]^{1/3} \frac{R_{AB}}{T} \quad \Rightarrow \quad R_{AB} \propto T^a, \quad a \approx \left[\frac{E_G}{4kT} \right]^{1/3}.$$

Application 1: the p - p chain

First (and rate-limiting) step: $p + p \rightarrow d + e^+ + \nu_e$.

- $E_G(pp) = 493$ keV.
- At solar-centre conditions, $T = 2 \times 10^7$ K $\Rightarrow kT = 1.7$ keV.
- Gamow peak: $E_0 = 4.2 kT = 7.2$ keV; half-width $\Delta/2 = 4.1$ keV.
- Tiny window compared to the $\sim$ MeV-scale Coulomb barrier — yet this is where essentially *all* pp fusion occurs.
- $S_{pp}(0) = 3.8 \times 10^{-22}$ keV b (weak interaction $\Rightarrow$ extremely small S) $\Rightarrow$ mean proton lifetime $\sim 9 \times 10^9$ yr — sets the Sun's main-sequence lifetime.
- Rate scales as $R_{pp} \propto T^4$ near solar conditions.

Application 2: the CNO cycle

Rate-limiting step in massive stars: $p + {}^{14}\text{N} \rightarrow {}^{15}\text{O} + \gamma$.

- Much higher Coulomb barrier: e.g. $E_G(p + {}^{12}\text{C}) = 35.5 \text{ MeV}$, vs. 0.66 MeV for $p + d$.
- Larger $E_G/kT \Rightarrow$ much steeper T -dependence: $a \approx [E_G/4kT]^{1/3}$ is large, giving $R \propto T^{17-18}$ near $T = 2 \times 10^7 \text{ K}$.
- This extreme sensitivity explains why the CNO cycle is negligible in the Sun (1.6% of luminosity) but dominates in stars only modestly hotter/more massive (e.g. Sirius A).
- ${}^{12}\text{C}$ acts as a catalyst — recycled, not consumed; relative isotopic abundances (${}^{12}\text{C} : {}^{13}\text{C} : {}^{14}\text{N} : {}^{15}\text{N}$) settle into equilibrium ratios set by their relative fusion rates.

Application 3: resonant reactions — triple- α

- The simple "slowly varying $S(E)$ " picture breaks down when a nuclear resonance falls inside, or near, the Gamow window.
- Triple- α process: ${}^4\text{He} + {}^4\text{He} \rightleftharpoons {}^8\text{Be}$ ($E_R = 91.8$ keV), then ${}^4\text{He} + {}^8\text{Be} \rightleftharpoons {}^{12}\text{C}^*$ (Hoyle state, 7.65 MeV excitation, 300 keV above threshold).
- Both steps are resonance-enhanced — the cross-section (and effectively $S(E)$) is sharply peaked rather than smooth.
- Equilibrium populations of ${}^8\text{Be}$ and ${}^{12}\text{C}^*$ follow directly from chemical-potential balance,

$$\frac{n_8}{n_4^2} \propto \exp\left[-\frac{(m_8 - 2m_4)c^2}{kT}\right],$$

i.e. still a Boltzmann/Gamow-type exponential, but now governed by a resonance energy rather than the smooth tunnelling exponent.

- Demonstrates that the Gamow-peak machinery is the *generic* energy-selection mechanism; resonances modify $S(E)$, not the underlying tunnelling framework.

Application 4: from the lab to the stars

- Strategy: measure $\sigma(E)$ (hence extract $S(E)$) at the lowest energies attainable with an accelerator, then extrapolate $S(E)$ down to the stellar Gamow-peak energy E_0 .
- Works well when $S(E)$ is smooth and the extrapolation range is modest.
- **Electron screening:** target/beam electrons reduce the effective Coulomb barrier in laboratory measurements, artificially enhancing $\sigma(E)$ at very low E relative to the bare-nucleus value relevant in a fully ionised stellar plasma — must be corrected for.
- **Resonance hunting:** a major goal of underground low-background facilities (e.g. LUNA) is to measure $S(E)$ as close as possible to astrophysical E_0 , directly, to avoid long extrapolations and missed resonances.

Application 5: advanced burning stages

- The same $E_G \propto Z_A Z_B$ scaling governs carbon, neon, oxygen and silicon burning in massive stars ($M \gtrsim 8M_\odot$), at progressively higher T and ρ .
- Higher $Z_A Z_B \Rightarrow$ much larger $E_G \Rightarrow$ much steeper T -sensitivity and much shorter burning timescales (years $\rightarrow$ a single day for Si-burning in a $25 M_\odot$ star).
- Above $T \sim 10^9$ K, the new ingredient is *photodisintegration* by the thermal photon bath, governed by Boltzmann factors $\exp(-Q/kT)$ analogous in spirit to the Gamow-peak Boltzmann factor.
- Burning terminates near the iron group ($A \approx 56$), the point of maximum binding energy per nucleon, beyond which no further energy can be extracted by fusion.

Summary of key results

- 1 Coulomb tunnelling probability: $P(E) \approx \exp[-2\pi\eta] = \exp[-(E_G/E)^{1/2}]$, with Sommerfeld parameter $\eta = Z_A Z_B e^2 / (4\pi\epsilon_0 \hbar v)$ and Gamow energy $E_G = (\pi\alpha Z_A Z_B)^2 2m_r c^2$.
- 2 Cross-section factorisation defines the astrophysical S-factor:
$$\sigma(E) = \frac{S(E)}{E} \exp[-(E_G/E)^{1/2}].$$
- 3 Folding the Gamow factor with the Maxwell–Boltzmann tail produces the **Gamow peak**:
 $E_0 = [E_G(kT)^2/4]^{1/3}$, width $\Delta = \frac{4}{3^{1/2}2^{1/3}} E_G^{1/6} (kT)^{5/6}$.
- 4 Resulting reaction rate $R_{AB} \propto S(E_0) T^{-2/3} \exp[-3(E_G/4kT)^{1/3}]$, with effective power-law index $a \approx (E_G/4kT)^{1/3}$.
- 5 Applications span the full life of a star: pp chain (T^4), CNO cycle (T^{17-18}), resonance-dominated triple- α , and the increasingly steep advanced burning stages up to the iron group.

References

- A.C. Phillips, *The Physics of Stars*, 2nd ed., Wiley (1994), Ch. 4.
- J.N. Bahcall, *Neutrino Astrophysics*, Cambridge University Press (1989).
- C. Rolfs and W. Rodney, *Cauldrons in the Cosmos*, University of Chicago Press (1988).
- D.D. Clayton, *Principles of Stellar Evolution and Nucleosynthesis*, University of Chicago Press (1983).